



Guideline

Peri-operative Medication Management

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	All areas	Medical, Pharmacy, Nursing

1. Contents

Scope.....	1
Introduction	1
Risk Statement	2
Practical Points	2
Abbreviations	2
Surgery Bleeding Risk*	3
Medication Guidance	3
1.0 Anticoagulants	3
2.0 Antiplatelets	5
3.0 Cardiovascular system medications	6
4.0 Diabetes medications	7
5.0 Neurological system medications	9
6.0 Analgesia medications.....	12
7.0 Hormonal medications	13
8.0 Immunosuppressive medications.....	14
9.0 Complementary medications	15
10.0 Other medications	15

Introduction

This guideline provides information on medication management in the pre- and peri-operative period. It is not an exhaustive list and is intended as a guide only. Final decision on medication management should be agreed upon by the anaesthetist and surgeon.

This document should be utilised in conjunction with SCGOPHG Medicines Management Policies and Procedures

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Practical Points

All oral medications which are not being withheld should be administered up to one hour prior to surgery with 30mL of water.

It is essential that certain medicines be continued throughout the perioperative period to prevent the relapse of the treated condition or to avoid the effects of drug withdrawal. If enteral administration or absorption is compromised, medications may need to be given by an alternative route or changed to an alternative drug with a similar action.

Some medications should be withheld peri-operatively; this may be due to their potential to interact with medications used in the peri-operative period and/or their impact on risk of surgical complications (such as bleeding or infection).

For combination medications that require the withholding of an individual component, a bridging script for the other components needs to be provided to the patient or pharmacy.

Abbreviations

PAC	Pre-Admission Clinic
Pre-op	Pre-operatively
Peri-op	Peri-operatively
Post-op	Post-operatively
Mgmt	Management
DOACs	Direct oral anticoagulants
VTE	Venous thromboembolism
ACS/AMI	Acute Coronary Syndrome/Acute Myocardial Infarct
DOS	Day of surgery
HTN	Hypertension
HoTN	Hypotension
HF	Heart failure
CrCl	Creatinine Clearance
CVD	Cardiovascular disease
VRII	Variable rate insulin infusion
ACH-ase	Anticholinesterase
CAMs	Complementary and Alternative Medicines
HBR	High Bleeding Risk
LBR	Low Bleeding Risk
MBR	Minimal Bleeding Risk
LMWH	Low Molecular Weight Heparin
BP	Blood Pressure
CI	Contraindication
NPG	Nursing Practice Guideline
DES	Drug Eluting Stent
PAD	Peripheral Artery Disease

Surgery Bleeding Risk*

High Bleeding Risk	Intracranial, spinal, major cancer, thyroid, eye posterior chamber, hepatobiliary, aneurysm, major orthopaedic, urologic, gynaecologic, bowel resection, vascular
Low Bleeding Risk	Therapeutic endoscopy, lap cholecystectomy, hernia repair, hand/foot surgery, endovascular procedures
Minimal Bleeding Risk	Cataract only, superficial skin, minor breast surgery, carpal tunnel

*List is not exhaustive

Medication Guidance

1.0 Anticoagulants

- Refer to [SCGOPHCG Guideline – Periprocedural Management of the Anticoagulants](#) for detailed information
- Patient related and surgical risk factors for thrombosis and bleeding should be assessed to determine an overall peri-procedural anticoagulation management plan for individual patients.
- Multiple teams may need to be involved – haematology, cardiology, neurology, surgical, anaesthetics
- Timing of recommencing DOACs post-operatively will depend on bleeding risk of surgery. VTE prophylaxis may be considered in patients with high risk of VTE until DOAC restarted

Heparins and warfarin		
Medication	Pre-op action	Comments
LMWH <i>enoxaparin, dalteparin</i>	Withhold ^(1, 2)	If <i>prophylaxis</i> withhold for ≥12h If <i>treatment</i> withhold for ≥24h
Unfractionated heparin	Withhold	If subcut <i>prophylaxis</i> : withhold for 2h If IV <i>therapeutic</i> infusion: withhold for 6h* *Consult treating specialist to confirm
warfarin	Withhold 4-5 days before surgery [^]	allow INR to drop <1.5 If high risk of thromboembolism, consider bridging therapy and refer to Anti-coagulation Services. [^] Can continue for minor surgery, discuss with surgeon

[^]Vascular procedures: withhold warfarin for 5 days if INR range 2.0-3.0 and for 7 days if INR range 3.0 to 4.0

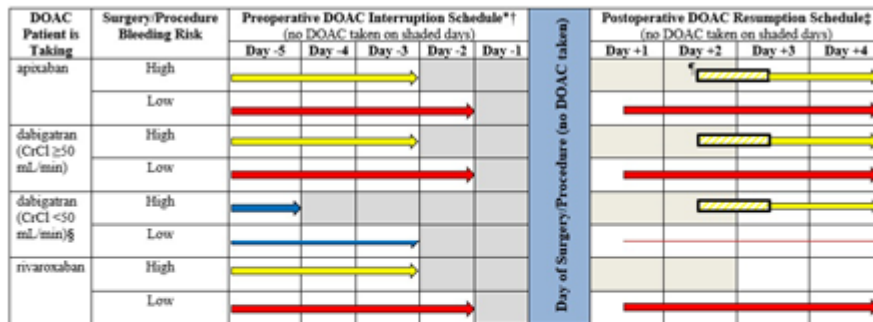
Direct Oral Anticoagulants					
Medication	Pre-op action				
	Renal function	Deep Regional* or Neuraxial Anesthesia	HBR	LBR	Superficial Regional / MBR
Assess BOTH need for regional anaesthesia and surgical bleeding risk					
apixaban ⁺ and rivaroxaban	≥50mLmin ⁻¹	Last dose 72h	Last dose 48h	Last dose 24h	Continue
	30-50mLmin ⁻¹	Last dose 72h	Last dose 72h	Last dose 48h	
	<30mLmin ⁻¹	Last dose >72h	Last dose >72h	Last dose >72h	
dabigatran	≥50mLmin ⁻¹	Last dose 72h	Last dose 72h	Last dose 48h	Continue

	30-50mLmin ⁻¹	Last dose 96h	Last dose 96h	Last dose 72h	
	<30mLmin ⁻¹	Last dose >120h	Last dose >120h	Last dose >120h	

* Deep regional blocks include deep cervical plexus, infraclavicular, paravertebral, quadratus lumborum, lumbar plexus and PENG blocks. Further information – Joint ESAIC/ESRA guidelines 2022. Clinical discretion advised on case-by-case basis. Figure 1 below has been taken from the PAUSE trial.

+ Apixaban is now finding use in patients with ESRF, on dialysis or with CrCl < 10 (usually 2.5mg BD). It is recommended to consult haematology for perioperative management of apixaban in this cohort of patients.

Figure 1. Perioperative DOAC Management Protocol



Legend:

*For patients at high bleeding risk, the 2 day (60-66 hour) interruption interval corresponds to 4-5 DOAC elimination half-lives and with an expected minimal-to-nil (<6%) residual anticoagulant level at the surgery/procedure.

†For patients at low bleeding risk, the 1 day (36-42 hour) interruption interval corresponds to 3-4 DOAC elimination half-lives and with an expected minimal (6-12%) residual anticoagulant level at the surgery/procedure.

‡For patients at low bleeding risk, DOAC was resumed within 24 hours post-surgery/procedure and for patients at high bleeding risk DOAC was resumed 48-72 hours post-surgery/procedure.

§Hatched segment of arrows refers to flexibility in the timing of DOAC resumption post-surgery/procedure, to account for inter-procedural variability in surgical haemostasis.

¶For patients on dabigatran with a CrCl <50 mL/min, the 2 day and 4 day interruption intervals account for the prolonged elimination half-life of dabigatran in patients with impaired renal function.

2.0 Antiplatelets

- Weigh the risk of thromboembolic events from stopping antiplatelet agents against the bleeding risk if continued
- It may be safe to continue antiplatelet agents before minor surgery with LBR. It may be necessary to reduce the antiplatelet effect before a surgery with HBR.
- For patients with coronary stents, **consider delaying** elective surgery until dual antiplatelet treatment is no longer required
- Involve relevant teams – haematology, cardiology, neurology, surgical, anaesthetics
- **Vascular patients:**
 - continue antiplatelets for all endovascular procedures (except spinal drain patients – stop all except aspirin)
 - consider stopping non-aspirin antiplatelets unless required for coronary DES
 - consider changing from prasugrel or ticagrelor to clopidogrel after discussion with cardiology and vascular teams

Antiplatelet agents		
Medication	Pre-op action	Comments
aspirin	Case-by-case decision. Continue unless specific surgical CI e.g. NIISWA, eye posterior chamber. <i>Consider discussing with surgical team and specialists if necessary.</i>	Continue, particularly in those with one or more coronary stents. Can withhold 7 days pre-op in patients with very HRB or bleeding consequences are significant. Pre-CABG: cont if <70 yrs, withhold DOS if >70 yrs
clopidogrel	Case-by-case decision. Usually withhold 5-10 days pre-op ⁽³⁻⁵⁾	Potent antiplatelet with few studies showing association with perioperative bleeding.
prasugrel	Case-by-case decision. Usually withhold 7-10 days pre-op ⁽³⁻⁶⁾	
ticagrelor	Case-by-case decision. Usually withhold 3-5 days pre-op ⁽³⁻⁶⁾	Ticagrelor reversibly inhibits P2Y ₁₂ . If antiplatelet action is not desired, manufacturers advise withholding 5 days pre-op, although it may be enough to stop 3 days. After 24h 57% of platelet function is restored. ⁽³⁾

It may be necessary to actively minimise antiplatelet effect before surgeries with HBR, however caution is required when withholding antiplatelet medication. Generally, for patients with coronary stents, consider delaying elective surgery until dual antiplatelet treatment is no longer required and seek a cardiology opinion before stopping any antiplatelet agents in patients with coronary stents. Caution withholding in patients with a coronary DES placed in the last 6-12 months or bare metal coronary stent placed in the last 2-4 weeks.

Current guidance for DES:

- If elective – can cease at 6 months
- For ACS/AMI – should continue for 12 months. Consideration of earlier cessation requires cardiology guidance.

For patients on vascular initiated antiplatelets – these can be withheld for a short period (1-2 weeks) if needed if the vascular procedure was not within the last 3 months. Contact vascular team for patients with vascular stents on antiplatelets.

Antiplatelet therapy for secondary prevention should be recommenced when safe to do so depending on the bleeding risks of the surgical procedure and patient characteristics.

3.0 Cardiovascular system medications

Cardiovascular medications			
Medication	Pre-op action		Comments
	Minor, day, intracranial, NIISWA	Major surgery	
α_1 antagonist <i>prazosin</i>	Continue	Continue	
α_2 antagonist <i>clonidine</i>	Continue	Continue	
ACEIs <i>ramipril, enalapril, perindopril</i>	Continue	Withhold	Risk of peri-op pervasive HoTN. Consider continuing for minor and intracranial procedures or if BP poorly controlled. Consider withholding on DOS for non-intracranial major surgery. May require individual mgmt plan ^(4, 5)
ARBs <i>candesartan, irbesartan, telmisartan</i>			
Anti-anginal <i>nitrates, nicorandil</i>	Continue	Continue	May precipitate chest pain if withheld
Antiarrhythmics <i>amiodarone, flecainide</i>	Continue ⁽⁴⁾	Continue	Inform anaesthetist on DOS if patient is taking amiodarone due to drug interactions (incl. patients who have ceased recently) ⁽⁶⁾
Beta-blockers <i>metoprolol, atenolol, bisoprolol, carvedilol</i>	Continue ^(4, 5)	Continue	Rebound HTN can occur if stopped abruptly. Monitor BP closely post-op. ⁽⁴⁾
Calcium channel blockers <i>amlodipine, nifedipine, diltiazem, verapamil</i>	Continue ^(4, 5)	Continue	Monitor BP (and heart rate with diltiazem / verapamil)
Cardiac glycosides <i>digoxin</i>	Continue ⁽⁴⁾	Continue	Continue EXCEPT in certain cardiac procedures such as ablations and L atrial appendage occlusions. Consult cardiologist or surgeon
Diuretics for HTN <i>hydrochlorothiazide, amiloride</i>	Continue	Withhold	Increased risk of peri-op HoTN if continued. May increase risk of peri-op hypokalaemia and HoTN ⁽⁴⁾
Diuretics for HF <i>furosemide, spironolactone</i>	Continue	Withhold	Consider withholding mane dose, risk of peri-op HoTN Loop diuretics may increase risk of peri-op hypokalaemia and spironolactone may be withheld due to hyperkalaemia risk. ⁽⁴⁾

Lipid-lowering medications		
Medication	Pre-op action	Comments
Statins <i>atorvastatin, simvastatin</i>	Continue	Vascular surgery Most patients with PAD should receive statins. In patients not treated, statins should ideally be initiated ≥ 2 weeks before intervention for maximal plaque-stabilising effects
Bile acid sequestrants <i>cholestyramine</i>	Continue	May interfere with absorption of some medications used in the peri-op period. ⁽⁴⁾
Ezetimibe	Continue	Risk and benefit of continuing unknown. ⁽⁴⁾
Fibrates	Continue	Increased risk of rhabdomyolysis ⁽⁴⁾

fenofibrate, gemfibrozil

4.0 Diabetes medications

- Medical history, medication history pre-op planning is essential for optimal mgmt
- Maintaining blood glucose levels between 6 -12 is associated with better peri-op outcomes
- Specialist referral to endocrinology or a diabetes educator may be required prior to surgery for pts with persistent BSLs >12mmol/L or recurrent hypoglycaemic episodes (BGL <4mmol/L) or HbA1c >9%
- Consider an IV insulin/glucose infusion in: All T1DM undergoing major surgery, T2DM undergoing major surgeries that develop hyperglycaemia (BGL >12mmol/L), poorly controlled diabetes and diabetics requiring emergency surgery.
- BSLs to be monitored per [NPG – Diabetes Patient Management](#)
- **Please refer to Appendix 1 for insulin management if patient administers both insulin and a GLP-1 analogue (and is therefore required to follow a prolonged clear fluid fast). Please also consider endocrinology input if clinically appropriate.**

Insulins and non-insulin injectables		
Medication	Pre-op action	Day of surgery
Basal Long Acting <i>Optisulin ®, Toujeo ®</i>	Reduce dose by 50% the evening before surgery.	50% of morning dose and check BSLs
Rapid and Short acting <i>Actrapid®, Novorapid ® Humalog ®</i>	Withhold when fasting	Withhold when fasting 50% of morning dose and check BSLs
Intermediate acting <i>Humulin NPH®, Protaphane®</i>	Reduce dose by 50% the evening before surgery.	50% of morning dose and check BSLs
Mixed <i>Humulin 30/70® Ryzodeg 70/30® Novomix 30/70® Mixtard 30/70®</i>	Reduce dose by 20% the evening prior to surgery. Withhold when fasting	Withhold when fasting 50% of morning of surgery. Check BSLs
Non-insulin injectables <i>GLP-1 analogues: exenatide, liraglutide, semaglutide, dulaglutide</i>	Continue. <i>Please alert anaesthetist that patient is using this medication. Patients are directed to follow an altered fasting protocol: refer to section 2.6 of the SCGOPHCG NPG Pre-Operative care.</i>	Continue. <i>Please alert anaesthetist that patient is using this medication. Patients are directed to follow an altered fasting protocol: refer to section 2.6 of the SCGOPHCG NPG Pre-Operative care.</i>
Subcutaneous continuous insulin infusion pump (SCIIP)	Change line and set site away from surgery. ⁽⁷⁾	Refer to endocrine to assess continuation of SCIIP. Do not suspend or stop SCIIP without an alternative. Continue basal rate as usual. If latest HbA _{1c} <6.5% or BSL <5mmol set basal rate to 80% of usual. ⁽⁷⁾ Check BSLs.

Oral hypoglycaemic agents		
Medication	Pre-op action	Day of surgery
Acarbose	Withhold DOS ⁽⁷⁾	<i>Morning surgery: Withhold if fasting Afternoon surgery: take mane dose if eating</i>
Biguanides <i>metformin</i>	Withhold DOS ⁽⁷⁾	Take as normal, some references state withhold day before and DOS, consult anaesthetist. ⁽⁷⁾ If eGFR < 60 ml/min

		Contrast media: Omit on the day of the surgery and for 48hrs post-op
DPP4 inhibitors <i>linagliptin, saxagliptin, sitagliptin</i>	Withhold DOS ⁽⁷⁾	Patients with T2DM commencing liver reduction diet (LRD): Consider stopping DPP4 inhibitors, with close monitoring of capillary blood glucose. ⁽⁶⁾
Sulfonylureas <i>glimepiride, gliclazide, glibenclamide, glipizide</i>	Withhold DOS ⁽⁷⁾	Patients with T2DM commencing LRD: Patients taking sulfonylureas are at increased risk of hypoglycaemia during the LRD, discontinue when the LRD commences. ⁽⁶⁾
SGLT2 inhibitors <i>dapagliflozin, empagliflozin</i>	Withhold 72h prior	Alert - Risk of ketoacidosis. See SCGHPHCG Guideline SGLT2 inhibitors and euglycaemic ketoacidosis
Thiazolidinediones <i>pioglitazone</i>	Withhold DOS ^(7, 8)	

5.0 Neurological system medications

Anticholinesterases		
Medication	Pre-op action	Comments
donepezil	Continue ⁽⁶⁾	Some guidelines suggest stopping pre-op. If stopping, donepezil requires a 2–3 week cessation. NB: An extended period without donepezil may cause a decline in cognitive function
galantamine		
rivastigmine		

Anticonvulsants		
Medication	Pre-op action	Comments
phenytoin, valproate, carbamazepine, lamotrigine, levetiracetam	Continue ⁽⁴⁾	
Muscle relaxant <i>baclofen</i>	Continue	

Antidepressants		
Medication	Pre-op action	Comments
Reversible MAOI <i>moclobemide</i>	Case-by-case decision	Reversible MAOIs (e.g. moclobemide) – withhold for 24h. ⁽⁴⁾ NB: MAOIs interact with certain agents, particularly sympathomimetics: ephedrine, metaraminol; opioids: pethidine
Irreversible MAOIs <i>phenelzine, tranylcypromine</i>	Case-by-case decision	Irreversible MAOIs (e.g. tranylcypromine) – Consult psychiatrist as needs to be withheld for 2 weeks. ⁽⁴⁾ NB: MAOIs interact with certain agents, particularly sympathomimetics: ephedrine, metaraminol; opioids: pethidine
mirtazapine	Continue ⁽⁴⁾	If stopping, taper for 1-2 weeks before surgery to minimize sleep disturbance ⁽⁴⁾
SSRIs e.g. fluoxetine, sertraline, paroxetine	Continue ⁽⁴⁾	Withdrawal can occur if stopped abruptly, consider tapering for CNS surgeries. ⁽⁴⁾ May increase bleeding risk due to potential antiplatelet effects. May increase risk of serotonin syndrome when used with other serotonergic agents (e.g. pethidine, tramadol, fentanyl)
SNRIs e.g. venlafaxine, duloxetine,		
TCAs e.g. amitriptyline, nortriptyline, dosulepin	Continue ⁽⁴⁾	Withdrawal can occur if stopped abruptly. If stopping, taper for 1-2 weeks before surgery to minimize sleep disturbance. Due to increased risk of arrhythmias and interactions with vasopressor drugs inform anaesthetist on the day of surgery if patient is taking a TCA

Antipsychotics		
Medication	Pre-op action	Comments
amisulpride, aripiprazole, quetiapine, olanzapine, chlorpromazine, haloperidol, risperidone, paliperidone	Continue ⁽⁴⁾	Continue unless prolonged QT interval pre-op. ⁽⁴⁾ Consult patient's psychiatrist if guidance required.
Clozapine	Withhold 12 hours prior	Consider checking clozapine levels. <i>Smoking Cessation</i> increases clozapine plasma levels regardless of whether electronic cigarettes or nicotine replacement therapy are used. Monitor plasma levels, as dose reduction (under the guidance of a Psychiatrist) may be necessary

Anxiolytics		
Medication	Pre-op action	Comments
buspirone	Continue ⁽⁴⁾	
benzodiazepines <i>diazepam, oxazepam</i>	Case-by-case decision – consult anaesthetist ⁽⁴⁾	Usually continue for chronic anxiety. ⁽⁴⁾ Additive effects or increased tolerance to peri-op anaesthetic and sedative agents may occur. Monitor for over-sedation.

Medications for Parkinsonism		
Medication	Pre-op action	Comments
Anticholinergics <i>benztropine</i>	Continue ⁽⁴⁾	Monitor for HoTN
Dopamine agonists <i>Bromocriptine, Sinemet®, pramipexole, Madopar®</i>	Continue ⁽⁴⁾	Withholding or missing doses can cause parkinsonian crisis. If oral route is unavailable, consider rotigotine patches or IV apomorphine (<i>for specialist use only</i>). Monitor for HoTN
MAO-B inhibitors <i>Selegiline</i>	Continue ⁽⁴⁾	Monitor for HoTN

Mood Stabilisers		
Medication	Pre-op action	Comments
lithium	Minor: continue ⁽⁴⁾ Major: Withhold 24h prior	Alert anaesthetist: lithium can prolong the effect of neuromuscular blockers. Risk of lithium toxicity if renal function deteriorates peri-op. Maintain hydration and avoid drugs that may cause peri-op renal impairment. Monitor serum levels ⁽⁴⁾
valproate	Continue ⁽⁴⁾	

Psychostimulants		
Medication	Pre-op action	Comments
dexamphetamine, lisdexamphetamine, methylphenidate	Consider withholding on morning of surgery.	Can lower seizure threshold, increase risk of HTN and arrhythmias and interact with peri-op medicines. ⁽¹⁾

Drugs of abuse		
Medication	Pre-op action	Comments
Alert anaesthetist if illicit drug use is suspected		

6.0 Analgesia medications

Analgesics		
Medication	Pre-op action	Comments
NSAIDs (Non-Selective) e.g. ibuprofen, naproxen, diclofenac	Case-by-case decision ⁽⁶⁾	Withhold for 2-3 days before surgery due to reversible antiplatelet effects. All NSAIDs may cause renal toxicity.
COX-2 (Selective) e.g. celecoxib, meloxicam	Case-by-case decision ⁽⁶⁾	May continue unless post-op renal impairment is a concern – minimal effects on platelet function.
Gabapentanoids pregabalin, gabapentin	Inform Anaesthetist Continue as usual	
Opioids morphine, oxycodone, codeine, methadone	Inform Anaesthetist Continue as usual	
Opioids (serotonergic) tramadol, pethidine	Inform Anaesthetist Continue as usual	
Community Program for Opioid Pharmacotherapy (CPOP)	Inform Anaesthetist Continue as usual	CPOP patients have complex pain management needs. See SCGH CPOP guideline for further information regarding the management of admitted CPOP patients

Opioid Receptor Antagonists		
Medication	Pre-op action	Action after surgery
naltrexone	Consider withholding 72h prior to surgery Consult anaesthetist/ pain physician	Note that patients are potentially very opioid sensitive once naltrexone is withheld. Aim to maximise regional anaesthesia where possible. Consider discussion with prescribing pain and addiction specialist.

7.0 Hormonal medications

- The decision to stop or continue oral contraceptives must balance the risk of unwanted pregnancy against the risk of thromboembolism. Consider the type of surgery and patient risk factors.
- If continuing hormone therapy, ensure patient receives adequate thromboprophylaxis if indicated
- Oral and injectable progesterone only methods of contraception are considered low risk unless other cardiovascular risk factors are present
- Consult surgeon/anaesthetist/specialist

Hormone Replacement Therapy		
Medication	Pre-op action	Comments
Conjugated oestrogens <i>e.g. Premarin®, Premia continuous®</i>	Consult surgeon / anaesthetist / specialist May continue with appropriate VTE prophylaxis ⁽⁶⁾	If high risk of VTE consider withholding tibolone for 4 weeks prior to surgery in patients with high risk of VTE ⁽¹⁾ Transdermal administration of oestrogen avoids first pass metabolism and thus has less effect on the coagulation factors than oral HRT. The risk associated with transdermal HRT given at standard therapeutic doses is no greater than the baseline population risk.
oestradiol <i>e.g. Climara®, Estraderm®, Estradot</i>		
oestriol <i>e.g. Ovestin®</i>		
tibolone <i>Livial®</i>		

Hormonal Antineoplastic Drugs		
Medication	Pre-op action	Comments
Selective Oestrogen Receptor Modulators (SERMs) <i>Tamoxifen</i>	Continue with appropriate VTE prophylaxis	Breast cancer prevention: consider withholding for 2 weeks prior to surgery for procedures with moderate to high risk of VTE. ⁽¹⁾ If patient is receiving active breast cancer treatment: discuss with oncologist. ⁽¹⁾ Note: risk of DVT is lower with aromatase inhibitors than with SERMs.
Aromatase Inhibitors <i>Anastrozole, letrozole, exemestane</i>		

Oral Contraceptive Pill (OCP)		
Medication	Pre-op action	Comments
Combined Oral Contraceptive <i>e.g. Estelle®, Yaz®, Norimin®</i>	Continue with appropriate VTE prophylaxis	If high risk of VTE consider withholding for 4 to 6 weeks prior to surgery in patients with high risk of VTE. ⁽¹⁾
Progesterone Only <i>Visanne®, Microlut®</i>		

8.0 Immunosuppressive medications

- Increased risk of surgical site infection and may delay wound healing.
- Consult surgeon/anaesthetist/specialist for peri-op planning if on high dose steroids pre-op

Corticosteroids (oral)		
Medication	Pre-op action	Comments
cortisone, dexamethasone, prednisolone	Case-by-case decision. Continue and consider periop stress dose supplementation. ⁽⁹⁾	Plan will vary depending on dose and surgery type. Patients with adrenal suppression due to corticosteroid use or taking corticosteroids for adrenal insufficiency require corticosteroid protection for adrenal crisis during and 24–48hrs after surgery. <i>NB: hydrocortisone 20mg is ≈ to prednisolone 5mg.</i>

Immunomodulators		
Medication	Pre-op action	Comments
Biologic response modifiers <i>e.g. infliximab, golimumab, adalimumab, etanercept, anakinra, rituximab</i>	Consider withholding . <i>Consult anaesthetist/ prescribing doctor.</i> If plan to withhold usually 1-2 cycles before surgery.	Restart once wounds have healed or 1-2 weeks after surgery. Monitor for signs of infection. Increased risk of peri-op infections with TNF-α antagonists.
Gout agents <i>allopurinol, colchicine</i>	Withhold colchicine on the morning of surgery. ⁽⁴⁾ Continue allopurinol	If plan to withhold – withhold on morning of surgery. <i>NB: risk of allopurinol toxicity with renal impairment.</i>
hydroxychloroquine, sulfasalazine	Continue	
methotrexate	Continue ⁽¹⁰⁾	If plan to withhold: 7 days pre-op
leflunomide	Case-by-case decision. Consult with prescribing doctor.	If plan to withhold: 11 days pre-op, consider chelation with cholestyramine
azathioprine	Continue ⁽¹⁰⁾	If plan to withhold: 7 days pre-op

9.0 Complementary medications

- Careful questioning is essential to obtain an accurate and complete medication history. Herbal medicines and supplements have the potential to cause adverse effects or interact with other medicines.
- Refer to NMHS Policy – [Complementary and Alternative Medicines](#)

Herbal and Vitamin Supplements		
Medication	Pre-op action	Comments
calcium	Continue	Check for medication interactions
colecalfiferol	Continue	Monitor renal function
magnesium	Continue	Check for medication interactions
All other supplements	Withhold for 2 weeks	If a patient is seen at PAC less than 1 week prior to surgery, advise to withhold the CAMs as soon as practicable and discuss with anaesthetist if required ⁽³⁾

10.0 Other medications

Asthma and COPD medications		
Medication	Pre-op action	Comments
β_2 -agonists <i>salmeterol, formoterol, indacaterol, terbutaline, salbutamol</i>	Continue ⁽⁴⁾	Administer morning of surgery
Inhaled corticosteroids (ICS) <i>fluticasone, ciclesonide, beclomethasone, budesonide</i>	Continue ⁽⁴⁾	Administer morning of surgery. Review patients on high doses i.e. >750microg fluticasone daily or >1500microg daily for other ICS ⁽⁹⁾ Adrenal insufficiency due to ICS is rare ⁽⁹⁾
Anticholinergics <i>ipratropium, tiotropium, umecclidinium</i>	Continue ⁽⁴⁾	
theophylline	Withhold from evening before surgery	Due to narrow therapeutic range has the potential to cause arrhythmias and neurotoxicity. There are interactions with some common perioperative medicines. ⁽⁴⁾
Leukotriene inhibitors <i>montelukast</i>	Continue ⁽⁴⁾	

Drugs for Benign Prostatic Hyperplasia (BPH)		
Medication	Pre-op action	Comments
Selective α -blockers <i>tamsulosin</i>	Continue	Ensure eye surgeon aware - intraoperative floppy iris syndrome has occurred during some eye surgeries ⁽⁴⁾
5- α -reductase inhibitors <i>dutasteride</i>	Continue ⁽⁶⁾	NB: CYP3A4 inhibitors (strong) may increase serum concentration

Glaucoma medications		
Medication	Pre-op action	Comments
β -blockers <i>timolol, betaxolol</i>	Continue	

Prostaglandin analogues <i>latanoprost, bimatoprost, travoprost</i>		
α_2 -agonists <i>brimonidine</i>		
Carbonic anhydrase inhibitors <i>brinzolamide, dorzolamide</i>		

H ₂ antagonists and Proton Pump Inhibitors		
Medication	Pre-op action	Comments
H ₂ antagonists <i>famotidine, nizatidine</i>	Continue ⁽⁴⁾	
Proton Pump Inhibitors <i>esomeprazole, lansoprazole, omeprazole, pantoprazole, rabeprazole</i>	Continue ⁽⁴⁾	

Osteoporosis medications		
Medication	Pre-op action	Comments
Bisphosphonates <i>alendronate, risedronate, zoledronic acid</i>	Withhold bisphosphonates on the morning of surgery due to need to administer with full glass of water and remain upright.	Osteonecrosis of the jaw has been associated with dental surgery and bisphosphonate use. Check with surgeon/ specialist for patients; requiring extensive bony surgery, a history of long-term bisphosphonate use, or who are also receiving chemotherapy or glucocorticoids. ⁽⁴⁾

Potassium supplements		
Medication	Pre-op action	Comments
potassium chloride	Continue	Monitor potassium levels. Continue if indicated.

Thyroid medications		
Medication	Pre-op action	Comments
levothyroxine	Continue ⁽⁴⁾	
liothyronine		
carbimazole		

Acknowledgements

- King Edward Memorial Hospital Pharmaceutical & Medicines Management Clinical Practice Guideline – Pre-operative Medication Management
- Fiona Stanley Fremantle Hospitals Group – Preoperative and preintervention medication management guideline
- Princess Alexandra Hospital - PAH Prescribing Guidelines

Related Documents

Legislation:

- [Statewide Medicines Formulary Policy MP 0077/18](#)
- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Health Services Act 2016

Authorising Policy and Standard/s:

- [Clinical Governance, Safety and Quality Policy Framework](#)
- NSQHS Standards:  see <https://www.safetyandquality.gov.au/our-work/assessment-to-the-nsqhs-standards/nsqhs-standards-second-edition/>

Standards, procedures, guidelines:

- Periprocedural Management of the Anticoagulants (CPG005) SCGHOPHCG
- SCGT2 inhibitors (glifozins) and euglycaemic ketoacidosis (MMG037) SCGHOPHCG
- SCGHOPHCG Nursing Practice Guideline – Pre-operative Care
- Nursing Practice Guideline – Diabetes Patient Management (NPG2) SCGHOPHCG
- Complementary and Alternative Medicines NMHS Policy
- Community program for opioid pharmacotherapy (CPOP) SCGH Medication Management Procedure

References

1. James D. Douketis ACS, Scott Kaatz, Richard C. Becker, Joseph A. Caprini, Andrew S. Dunn, David A. Garcia, Alan Jacobson, Amir K. Jaffer, David F. Kong, Sam Schulman, Alexander G.G. Turpie. Perioperative Bridging Anticoagulation in Patients with Atrial Fibrillation receiving a direct oral anticoagulant. JAMA intern Med. 2019;179(11):1469-78.
2. David Keeling RCT, Henry Watson. Peri-operative management of anticoagulation and antiplatelet therapy. British Journal of Haematology. 2016;175(4):602-13.
3. Roberta Rossini GM, Luigi Oltrona Visconti, Ezio Bramucci, Battistina Castiglioni, Stefano De Servi, Corrado Lettieri, Maddalena Lettino, Emanuela Piccaluga, Stefano Savonitto, Daniela Trabattini, Davide Capodanno, Francesca Buffoli, Alessandro Parolari, Gianlorenzo Dionigi, Luigi Boni, Federico Biglioli, Luigi Valdatta, Andrea Droghetti, Antonio Bozzani, Carlo Setacci, Paolo Ravelli, Claudio Crescini, Giovanni Staurengi, Pietro Scarone, Luca Francetti, Fabio D'Angelo, Franco Gadda, Andrea Comel, Luca Salvi, Luca Lorini, Massimo Antonelli, Francesco Bovenzi, Alberto Cremonesi, Dominick J Angiolillo, Giulio Guagliumi. Perioperative management of antiplatelet therapy in patients with coronary stents undergoing cardiac and non-cardiac surgery: a consensus document from Italian cardiological, surgical and anaesthesiological societies. EuroIntervention. 2014;10(1):38-46.
4. Visala Muluk SLC, Christopher Whinney. Perioperative Medication Management. UpToDate2023.
5. Sary Aranki SM, Hadi D Toeg. Coronary artery bypass surgery: Perioperative medical management. UpToDate2023.
6. Association UCP. The Handbook of Perioperative Medicines2023.
7. Nadia A Khan WAG, Enrico Cagliero. Perioperative management of blood glucose in adults with diabetes mellitus. UpToDate2023.
8. Ireland AoAogBa. Peri-operative management of the surgical patient with diabetes Anaesthesia. 2015;70:1427-40.
9. A Hamrahian SR, S Milan. The management of the surgical patient taking glucocorticoids. UpToDate2023.

10. Axford J. Preoperative evaluation and perioperative management of patients with rheumatic diseases. UpToDate2023.

Authorisation & Version Control

Policy Sponsor	SCGOPHCG Director of Clinical Services				
Policy Contact	SCGOPHCG Medicines and Therapeutics Committee Executive Officer				
Date First Issued:	17/07/2023	Last Reviewed:	21/11/2025	Review Date:	17/07/2026
Version No.	1.3	Reference #:	MMG051		
Approved by:	SCGOPHCG Medicines and Therapeutics Committee			Date:	28/11/2025
Endorsed by:	SCGOPHCG Medicines and Therapeutics Committee			Date:	28/11/2025
Acknowledgments:	Head of Department - Anaesthesia, OPH Pharmacy, SCGH Pharmacy, Diabetes and Endocrinology, KEMH Pre-Admission Clinic				
Standards Applicable	Relevant NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1. Guideline for patients on Insulin and GLP-1 analogues

Insulin Regimen	Instructions
Basal-Bolus (e.g. Optisulin® with Novorapid®)	- Halve the basal (long-acting) dose the night prior to the procedure - Avoid bolus (short-acting) doses of insulin on the clear fluid fast day and day of procedure unless advised otherwise - Manage blood sugars with carbohydrate-rich or non-carbohydrate clear fluids as appropriate
Mixed insulin (e.g. Novomix®, Ryzodeg®)	- Halve the dose on the clear fluid fast day - Replace meals and administer dose(s) with carbohydrate-rich clear fluids - Outside of meal times, maintain hydration with non-carbohydrate clear fluids
Insulin pump	- Requires individualised plan. Consult anaesthetics; input from diabetes specialist may be required.

Guideline for patients on Insulin and GLP-1 analogues

Patients on GLP-1 analogues must fast on clear fluids only for 24 hours before surgery

This guideline outlines how to adjust insulin regimens during this period. Please also consider referring to endocrinology for further advice if clinically appropriate.

For more information on fasting guidelines speak to a member of staff or **scan the QR code**

